

SOFTWARE PROCESS & PROJECT MANAGEMENT



SOFTWARE PROCESS & SOFTWARE PROJECT MANAGEMENT

- **Software Processes**, which define **how** software is developed.
- **Software Project Management**, which ensures **that development activities are planned, organized, monitored, and controlled** to meet project goals.
- Software processes provide the **technical framework** for software development, while project management provides the **managerial framework** to ensure that these processes are executed efficiently.
- A project with an excellent process but poor management may exceed its budget or miss deadlines, whereas effective management without a well-defined process may produce low-quality software.



RELATIONSHIP BETWEEN SOFTWARE PROCESSES AND PROJECT MANAGEMENT

Customer Requirements



Software Project Management (Planning and Control Activities)



Software Development Process (Requirements → Design →
Coding → Testing)



High-Quality Software Product



Achievement of Project Objectives

Software Process

Defines development activities

Specifies technical workflow

Focuses on product quality

Standardizes development

Ensures repeatability

Software Project Management

Plans and manages those activities

Specifies schedule and resources

Focuses on project success

Coordinates people and resources

Ensures timely delivery



ACHIEVING PROJECT OBJECTIVES THROUGH SOFTWARE PROCESSES AND PROJECT MANAGEMENT

- The success of a software project is measured using five primary objectives.

Project Objective	Contribution of Software Process	Contribution of Project Management
Scope	Defines required development activities	Controls scope changes
Time	Defines sequence of tasks	Plans and monitors schedule
Cost	Supports effort estimation	Controls project budget
Quality	Includes verification and validation	Monitors quality metrics
Customer Satisfaction	Produces reliable software	Ensures stakeholder communication and timely delivery

CONSEQUENCES OF POOR INTEGRATION

Situation

Result

Good software process but poor project management

High-quality software delivered late or over budget

Good project management but poor software process

Project completed on time but software contains defects or fails to meet user requirements

Poor software process and poor project management

High likelihood of project failure

Good software process and good project management

High-quality software delivered successfully within project constraints



KEY TAKEAWAYS

- **Software Process** answers "**How will the software be developed?**"
- **Software Project Management** answers "**How will the project be planned, managed, and controlled?**"
- A software process provides the **technical roadmap**, while project management provides the **managerial roadmap**.
- **Successful software projects require both a well-defined software process and effective project management to achieve organizational and customer objectives.**

THE CAPABILITY MATURITY MODEL (CMM)



WHY DO SOFTWARE PROCESSES NEED IMPROVEMENT?

- Software Process is Dynamic
- A software process is **not permanent**.
- As technology, customer expectations, and development practices evolve, organizations must continuously improve the way they develop software.

GOALS OF PROCESS IMPROVEMENT

- Produce higher-quality software
- Reduce development cost
- Reduce software defects
- Improve productivity
- Deliver software faster
- Increase customer satisfaction
- Improve predictability of projects

WHY CAN'T WE KEEP USING THE SAME PROCESS?

- Frequent software failures
- High maintenance cost
- Poor communication
- Missed deadlines
- Customer complaints
- Difficult project tracking
- Inconsistent quality

WHAT IS CMM?

1. THINK!

Suppose you and your friends are asked to build a software project.

How will you start?

- Start coding immediately.
- Discuss requirements.
- Divide work.



2. IMAGINE THIS...



Five software companies are asked to develop the same Online Shopping App.

Will all companies produce software of the same quality?

Answer: **No!**

3. WHY NOT?



Company A

- No planning
- No documentation
- Starts coding immediately



Company B

- Good planning
- Standard process
- Proper testing



Company C

- Measures quality
- Uses metrics
- Continuously improves

→ Every company follows a different process, so their results differ.

4. BIG QUESTION

How do we know which company has the better software development process?

That's exactly why CMM was created!

5. WHAT IS CMM?

Capability Maturity Model (CMM) is a framework that tells us how mature, organized and effective a software company's development process is.

REMEMBER THIS!

CMM is like a report card for a software company's development process.

IMPORTANT NOTE

CMM does **NOT** measure how good the software is.

CMM measures how good the software development **PROCESS** is.

6. WHY IS CMM USED?

Companies use CMM to:

- ✓ Improve software quality
- ✓ Reduce bugs
- ✓ Finish projects on time
- ✓ Reduce costs
- ✓ Improve customer satisfaction
- ✓ Standardize software development
- ✓ Continuously improve

FINAL TAKEAWAY

Good software is not created by luck – it is created by following a good software development process. CMM helps organizations build that process.

7. REAL-LIFE EXAMPLE

Hospital A



- No records
- Random medicines
- No schedule
- No quality checks

Hospital B



- Standard procedures
- Patient records
- Regular monitoring
- Quality checks

Which hospital would you trust?

Obviously **Hospital B!**

CAPABILITY MATURITY MODEL (CMM)

Capability Maturity Model (CMM) is a framework developed by the **Software Engineering Institute (SEI)** to help organizations improve their software development processes in a systematic and gradual manner.

Purpose

- Measure process maturity
- Guide process improvement
- Improve quality
- Improve productivity
- Reduce project risks

CAPABILITY MATURITY MODEL.

Software Process Capability

Process Capability refers to the expected level of performance that an organization can achieve using its current software process.

Capability = What results can we expect?

Process Maturity

Process Maturity represents how disciplined, standardized, and well-controlled a software process is.

Maturity = How well organized is our process?



A simple way to express project capability is:

$$\text{Project Capability} = \frac{\text{Available Resources}}{\text{Required Resources}} \times 100\%$$

where resources may be measured in terms of effort (hours), budget, or personnel.

Example

Suppose a software company has:

- Available productive hours = 600 hours
- Project effort required = 500 hours

Then,

$$\text{Project Capability} = \frac{600}{500} \times 100 = 120\%$$

If instead:

- Available hours = 450
- Required hours = 500

then

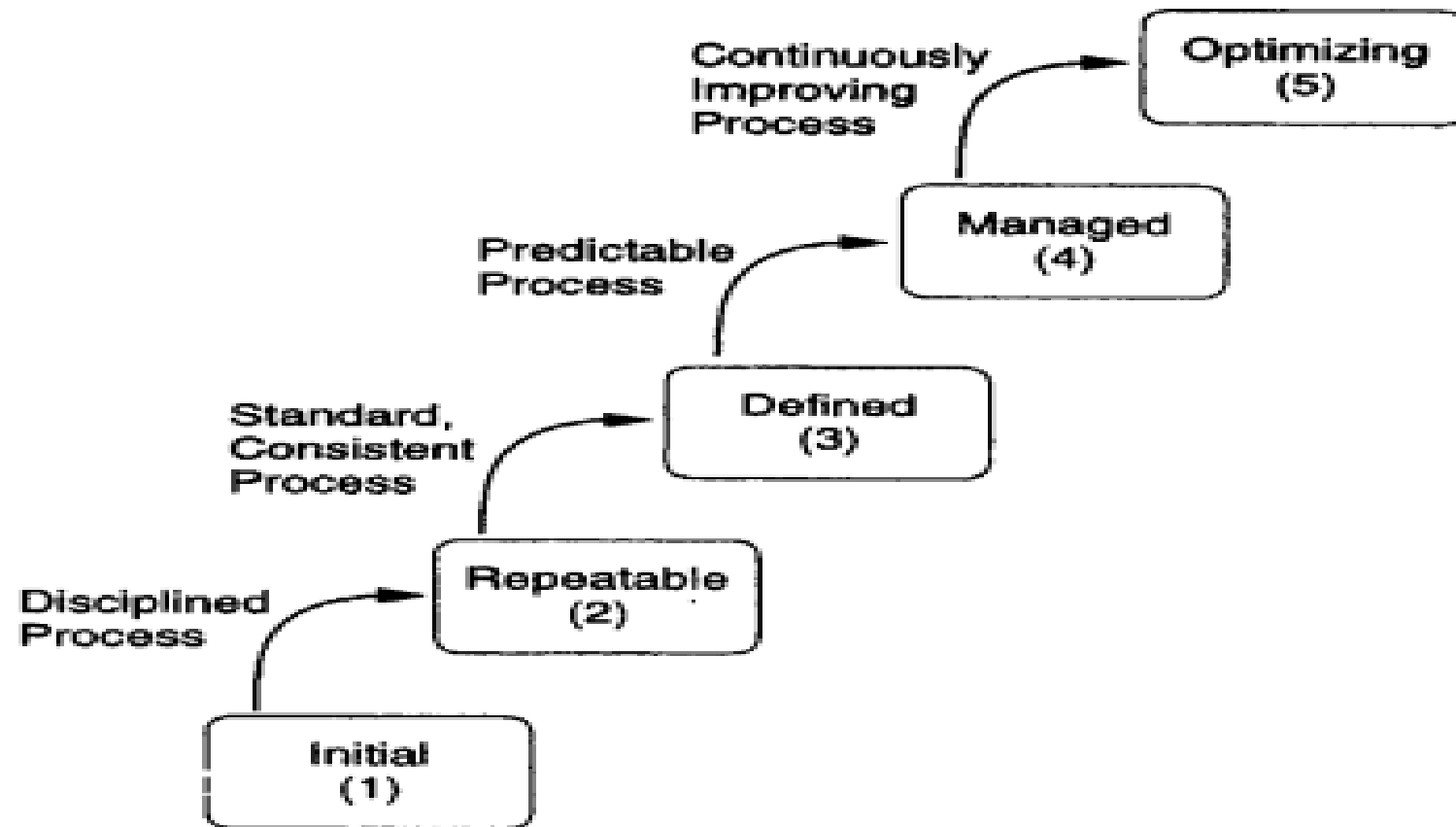
$$\frac{450}{500} \times 100 = 90\%$$

This indicates the project has only 90% capability, suggesting that additional resources, a longer schedule, or reduced scope may be required.

RELATIONSHIP BETWEEN CAPABILITY AND MATURITY

Project Capability	Project Maturity
Measures the ability to deliver a project	Measures the quality of the organization's project management processes
Focuses on available resources and capacity	Focuses on process effectiveness and consistency
Answers: "Can we do the project?"	Answers: "How well can we manage the project?"
Measured using hours, budget, staff, or equipment	Assessed using maturity frameworks such as CMM or CMMI
Can change from project to project	Usually changes gradually as organizational processes improve

CAPABILITY MATURITY MODEL.



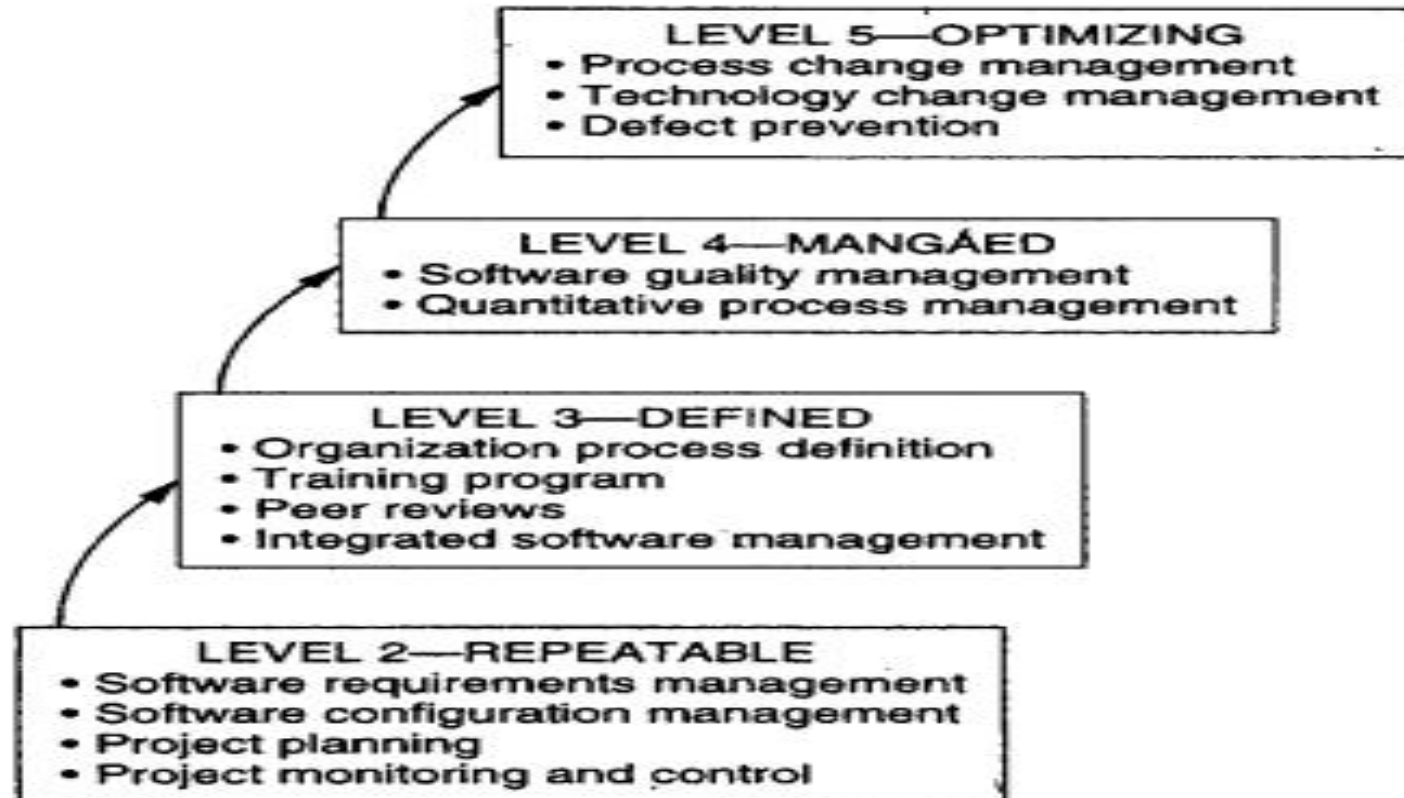
A SIMPLE MNEMONIC FOR STUDENTS

A very easy mnemonic to remember the five levels is:

- **I** – Initial
- **R** – Repeatable
- **D** – Defined
- **M** – Managed
- **O** – Optimizing

"I Really Develop Mature Organizations." (I – R – D – M – O)

KEY PROCESS AREAS



LEVEL I – INITIAL

- Ad hoc development
- No standard process
- Success depends on individuals
- Poor planning
- Poor documentation
- Frequent schedule delays
- Frequent cost overruns

LEVEL 2 – REPEATABLE

- Basic project management practices are introduced.
- Organizations begin to:
- Plan projects
- Estimate cost
- Track schedules
- Manage requirements
- Maintain configuration control

LEVEL 2 KEY PROCESS AREAS

- Requirements Management
- Project Planning
- Project Monitoring
- Configuration Management
- Quality Assurance

Benefits

- Better planning
- Controlled development
- Fewer surprises

LEVEL 3 – DEFINED

- The organization develops a **standard software process** that everyone follows.
- Standard documentation
- Standard coding guidelines
- Standard review process
- Formal testing process
- Employee training

LEVEL 3 KEY PROCESS AREAS

- Organization Process Definition
- Training Program
- Peer Reviews
- Integrated Software Management

Benefits

- Consistency
- Better communication
- Easier employee training

LEVEL 4 – MANAGED

- Now the organization begins measuring everything.

Examples of Metrics

- Defect Density
- Productivity
- Testing Coverage
- Project Cost
- Schedule Variance

LEVEL 4 KEY PROCESS AREAS

- Quantitative Process Management
- Software Quality Management

Benefits

- Predictable quality
- Predictable schedules
- Better risk management

LEVEL 5 – OPTIMIZING

- The organization continuously improves itself.

Activities

- Defect Prevention
- Process Improvement
- Technology Adoption
- Automation
- Root Cause Analysis
- Organizations continuously learn from previous projects.

LEVEL 5 KEY PROCESS AREAS

- Process Change Management
- Technology Change Management
- Defect Prevention

Benefits

- Continuous innovation
- Reduced defects
- Lower costs
- Better customer satisfaction

A company is developing a payroll system.

Available resources

- Budget = \$100,000
- Required budget = \$90,000

Project capability:

$$\frac{100,000}{90,000} \times 100 = 111.1\%$$

The company has enough financial resources to undertake the project.

However, consider two different organizations:

Company A

- CMM Level 1
- No standard development process
- Frequent requirement changes
- Poor documentation
- Many software defects

Although it has **111% capability**, the project may still fail because the organization lacks mature processes.

Company B

- CMM Level 5
- Standardized processes
- Continuous monitoring
- Automated testing
- Effective risk management

With the same **111% capability**, Company B is far more likely to deliver the project on time, within budget, and with higher quality because of its greater process maturity.

Scenario: A software development company has 5 developers, each working 20 days per month at 8 hours per day, giving a gross capacity of 800 hours. A project requires 620 hours of effort. The organization's productive utilization varies according to its Capability Maturity Model (CMM) level as follows:

CMM Level Productive Utilization

Level 1 – Initial 60%

Level 2 – Managed 70%

Level 3 – Defined 75%

Level 4 – Quantitatively Managed 82%

Level 5 – Optimizing 88%

- a) Calculate the effective project capability (in hours) for each CMM level.
- b) Calculate the capability ratio for each level
- c) Determine whether the project can be completed successfully at each maturity level.
- d) Explain how increasing CMM maturity improves project capability, project predictability, and software quality.

Step 1: Calculate gross capacity

Gross Capacity = $5 \times 20 \times 8 = 800$ hours

Step 2: Calculate effective project capability

Developers do not spend all their time coding because of meetings, reviews, support, and administration.

Effective Capability = $800 \times 75\% = 600$ productive hours

Therefore, the team's capability for the month is: 600 productive hours

Step 3: Compare capability with planned workload

Suppose the project contains the following work:

Activity	Estimated effort
Requirements analysis	70 hours
System design	80 hours
Development	320 hours
Testing	110 hours
Deployment	40 hours
Total required effort	620 hours

Step 4: Calculate the capability ratio

$$\text{Capability Ratio} = \frac{\text{Available Capability}}{\text{Required Effort}}$$

$$= \frac{600}{620} = 0.968$$

$$\text{Capability Percentage} = 0.968 \times 100 = 96.8\%$$

Interpretation

The team can complete approximately:

96.8%

of the planned work during the month.

The capacity shortfall is:

$$620 - 600 = 20 \text{ hours}$$

Therefore, the project manager may:

- reduce the project scope by about 20 hours;
- add temporary resources;
- authorize overtime; or
- extend the schedule.

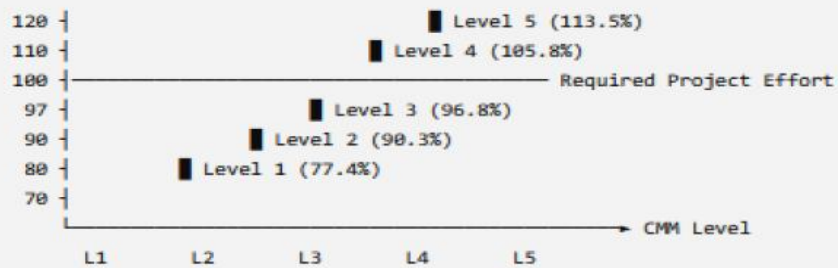
Final conclusion

The project requires **620 hours**, but the team has an effective capability of only **600 hours**. Therefore, the project is currently **under-capacity by 20 hours**, or approximately **3.2%**.

CMM Level	Maturity Description	Productive Utilization	Effective Capability (Hours)	Capability Ratio	Project Outcome
Level 2 – Managed	Basic project management practices exist	70%	560	$560/620 = 90.3\%$	Most work completed, some schedule slippage
Level 3 – Defined	Standardized organizational processes	75%	600	$600/620 = 96.8\%$	Minor shortfall; project largely on schedule
Level 4 – Quantitatively Managed	Processes measured and controlled	82%	656	$656/620 = 105.8\%$	Project completed on time with contingency
Level 5 – Optimizing	Continuous process improvement	88%	704	$704/620 = 113.5\%$	Early delivery with capacity for innovation

Graphical Interpretation

Capability (%)



CMM Level 1 (Initial)

Gross Capacity:

$$5 \times 20 \times 8 = 800 \text{ hours}$$

Effective Capability:

$$800 \times 0.60 = 480 \text{ hours}$$

Capability Ratio:

$$\frac{480}{620} = 0.774 = 77.4\%$$

Interpretation: Only about 77% of the required work can be completed due to poor planning, frequent rework, and inconsistent processes.

CMM Level 3 (Defined)

Effective Capability:

$$800 \times 0.75 = 600 \text{ hours}$$

Capability Ratio:

$$\frac{600}{620} = 0.968 = 96.8\%$$

Interpretation: Standardized development and management processes reduce inefficiencies, allowing the team to complete nearly all planned work.

CMM Level 5 (Optimizing)

Effective Capability:

$$800 \times 0.88 = 704 \text{ hours}$$

Capability Ratio:

$$\frac{704}{620} = 1.135 = 113.5\%$$

Interpretation: Continuous improvement, automation, and defect prevention enable the organization to exceed the required effort, leaving spare capacity for innovation, training, or additional features.

In this example, effective capability rises from 480 hours (77.4%) at CMM Level 1 to 704 hours (113.5%) at CMM Level 5, demonstrating how higher maturity translates into greater project performance, predictability, and delivery success.

SUMMARY

Level	Process	Main Focus	Organization Behavior
Initial	Unpredictable	Survival	Depends on individuals
Repeatable	Basic	Project Management	Repeat successful projects
Defined	Standardized	Organization-wide Process	Everyone follows same process
Managed	Measured	Metrics	Data-driven decisions
Optimizing	Continuously Improving	Innovation	Constant improvement

ADVANTAGES OF CMM

- Better software quality
- Reduced defects
- Better project planning
- Better cost estimation
- Predictable project delivery
- Continuous process improvement
- Higher customer satisfaction
- Better teamwork

LIMITATIONS OF CMM

- Expensive to implement
- Requires organizational commitment
- Large amount of documentation
- Time-consuming
- Difficult for small organizations
- Focuses more on process than technical innovation

CASE STUDY: PROJECT MANAGEMENT PRACTICES AT INFOSYS AND THEIR PROCESS MATURITY

INTRODUCTION

- Infosys is one of India's largest IT services and consulting organizations, delivering software development, maintenance, cloud, AI, and digital transformation projects worldwide.
- The company is widely recognized for its disciplined project management framework and high process maturity standards.
- Infosys has consistently adopted industry best practices such as Capability Maturity Model Integration (CMMI), ISO standards, quantitative project management, and continuous process improvement to ensure predictable project outcomes

BACKGROUND

- Founded: 1981
- Headquarters: Bengaluru, India
- Industry: Information Technology Services and Consulting
- Employees: Over 300,000 globally
- Services:
 - Software Development
 - Application Maintenance
 - Cloud Computing
 - Data Analytics
 - Artificial Intelligence
 - Digital Transformation

PROJECT MANAGEMENT FRAMEWORK AT INFOSYS

Business Requirement Collection

- **Project Initiation**
- Requirement elicitation
- Scope definition
- Feasibility assessment
- Stakeholder identification
- Risk assessment

Infosys business analysts meet stakeholders and identify:

- Online banking requirements
- Mobile banking features
- Security requirements
- Regulatory compliance requirements

Stakeholder Identification

Stakeholder	Responsibility
Bank CIO	Sponsor
Business Users	Requirements
Infosys Project Manager	Delivery
Architects	Solution Design
Quality Team	Verification

- **b) Project Planning**
- **Work Breakdown Structure (WBS)**
- Resource allocation
- Cost estimation
- Schedule planning
- Quality planning

Feasibility Analysis

Deliverables

- Project Charter
- Initial Scope Statement
- Stakeholder Register
- Risk Register
- **Work Breakdown Structure (WBS)**

Resource Plan

Role	Count
Project Manager	1
Architects	2
Developers	30
Test Engineers	10
DevOps Engineers	3
Business Analysts	4

Risk Planning

Risk	Probability	Mitigation
Requirement Changes	High	Change Control Board
Data Breach	Medium	Security Reviews
Schedule Delay	Medium	Buffer Allocation
Resource Attrition	Low	Backup Resources

Schedule Baseline

Phase	Duration
Requirements	2 Months
Design	1 Month
Development	5 Months
Testing	2 Months
Deployment	1 Month
Closure	1 Month

- **Project Execution**
- Agile and Waterfall methodologies
- Sprint planning
- Regular status reviews
- Configuration management
- **d) Monitoring and Control**
- Earned Value Management (EVM)
- Defect tracking
- Risk monitoring
- Schedule variance analysis
- Effort variance analysis

Sprint-Based Development

Sprint 1

Features Delivered:

- User Registration
- Login Module
- Authentication Framework

Sprint 2

Features Delivered:

- Account Summary
- Statement Viewing
- Transaction History

Sprint 3

Features Delivered:

- Fund Transfers
- Beneficiary Management
- Bill Payments

Quality Activities

Throughout development:

- Peer Reviews
- Static Code Analysis
- Unit Testing
- Integration Testing
- Security Testing

Schedule Metrics

Metric	Planned	Actual
Sprint Completion	85%	83%
Tasks Completed	120	115

Quality Metrics

Metric	Value
Defects Found	35
Defects Resolved	32
Defect Density	0.4/KLOC

Cost Metrics

Metric	Value
Planned Cost	\$2.1 M
Actual Cost	\$2.0 M

- **e) Project Closure**
- Customer acceptance
- Lessons learned
- Knowledge repository updates
- Project performance review

Final Review

Project Performance

Customer Satisfaction Survey

Parameter	Target	Actual
Duration	12 Months	12 Months
Budget	\$5.0 M	\$4.9 M
Quality Goal	98%	98.2%

Category	Score
Quality	4.8/5
Communication	4.7/5
Delivery	4.9/5

PROCESS MATURITY AT INFOSYS

- **Characteristics of Level 5 Maturity**
- Process standardization across projects
- Quantitative project management
- Defect prevention mechanisms
- Continuous process improvement
- Predictive analytics for project outcomes

KEY PROJECT MANAGEMENT PRACTICES

Risk Management

Risk	Mitigation
Requirement changes	Change control board
Schedule slippage	Buffer planning
Resource attrition	Backup resource pool
Technology uncertainty	Proof of concept

Metrics-Driven Management

Project managers regularly monitor:

- Schedule Performance Index (SPI)
- Cost Performance Index (CPI)
- Defect Density
- Customer Satisfaction Index
- Productivity Metrics

- **Knowledge Management**

- Lessons learned from completed projects are captured and shared across teams.
- Benefits:
 - Reduced rework
 - Faster onboarding
 - Better estimation accuracy
 - Organizational learning

Quality Assurance

Quality is built into every development phase through:

- Peer reviews
- Code inspections
- Automated testing
- Defect prevention programs
- Root cause analysis



PROJECT CLOSURE

Final Review

Project Performance

Customer Satisfaction Survey

Process Maturity Mapping (CMMI Perspective)

Project Activity	Maturity Evidence
Standard Planning Templates	Level 3
Organization-wide Processes	Level 3
Metrics-based Management	Level 4
Statistical Estimation Models	Level 4
Defect Prevention Programs	Level 5
Continuous Process Improvement	Level 5
Knowledge Management	Level 5